

Diseño de un controlador para un sistema de segundo orden

→ Ecuaciones principales

$$V_e(t) = R i_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) - i_2(t)]$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) - i_2(t)] = R i_2(t) + R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$V_s = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

→ Transformada de Laplace

$$V_e(s) = R I_1(s) + L_s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)]$$

$$L_s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)] = R I_2(s) + R I_2(s) + \frac{I_2(s)}{C_s}$$

$$V_s(s) = R I_2(s) + \frac{I_2(s)}{C_s}$$

→ Procedimiento algebraico

$$V_e(s) = (L_s + 2R) I_1(s) - (L_s + R) I_2(s)$$

$$I_1(s) = \frac{1}{L_s + R} (L_s + \frac{1}{C_s} + 3R) I_2(s)$$

$$V_s(s) = (R + \frac{1}{C_s}) I_2(s)$$

$$\frac{V_s(s)}{V_e(s)} = \frac{R + \frac{1}{C_s}}{(L_s + 2R) \left(\frac{1}{L_s + R} (L_s + \frac{1}{C_s} + 3R) I_2(s) \right) - (L_s + R) I_2(s)}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{\frac{CR_s + 1}{C_s} I_2(s)}{(L_s + 2R) \left(\frac{CL_s^2 + 3CR_s + 1}{CL_s^2 + CR_s} I_2(s) \right) - (L_s + R) I_2(s)}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{\frac{CR_s + 1}{C_s} I_2(s)}{((L_s + 2R) \left(\frac{CL_s^2 + 3CR_s + 1}{CL_s^2 + CR_s} \right) - (L_s + R)) I_2(s)}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{\frac{CR_s + 1}{C_s} I_2(s)}{\frac{3CLRs^2 + (5CR^2 + L)s + 2R}{CLs^2 + CR_s} I_2(s)}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{(CR_s + 1)(L_s + R)}{3CLRs^2 + (5CR^2 + L)s + 2R}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{CLRs^2 + (CR^2 + L)s + R}{3CLRs^2 + (5CR^2 + L)s + 2R}$$

→ Error en estado estacionario

$$\lim_{s \rightarrow 0} s R(s) [1 - G(s)]$$

$$\lim_{s \rightarrow 0} \frac{CLR(0)^2 + (CR^2 + L)(0) + R}{3CLR(0)^2 + (5CR^2 + L)(0) + 2R} = \frac{R}{2R} = \frac{1}{2} = 0.5$$

$$e_{ss} = 0.5 = 50\%$$

→ Estabilidad en lazo abierto

$$R = 5 \text{ k}\Omega$$

$$L = 10 \mu\text{H}$$

$$C = 470 \mu\text{F}$$

$$\text{Denominador} = 3CLR s^2 + (5CR^2 + L)s + 2R$$

$$s_{1/2} = \frac{-(5CR^2 + L) \pm \sqrt{(5CR^2 + L)^2 - 4(3CLR)(2R)}}{2(3CLR)}$$

$$s_{1/2} = \frac{-(5(470 \times 10^{-6})(5 \times 10^3)^2 + 10 \times 10^{-6}) \pm \sqrt{(5(470 \times 10^{-6})(5 \times 10^3)^2 + 10 \times 10^{-6})^2 - 4(3(470 \times 10^{-6})(10 \times 10^{-6})(5 \times 10^3))}}{6((470 \times 10^{-6})(10 \times 10^{-6})(5 \times 10^3))}$$

$$s_1 = -414.947672 \text{ EG}$$

$$s_2 = -418.38566 \text{ EG}$$

s_1 y s_2 son menores a 0 $\therefore$ Es estable sobreamortiguada

